

DAY 6, AUGUST 29 & 30, UNIT 1

Agenda:

1. 1.1 Intro to Maps
2. Map Gallery Walk
3. HW: Ch. 1 pages 11-17 and complete Key Q #3 Questions



1.1 Introduction to Maps

Objective: Identify types of maps, the types of information presented in maps, different kinds of spatial patterns and relationships portrayed in maps

Essential Knowledge: Types of maps include reference maps and thematic maps. Types of spatial patterns represented on maps include absolute and relative distance and direction, clustering, dispersal, and elevation.



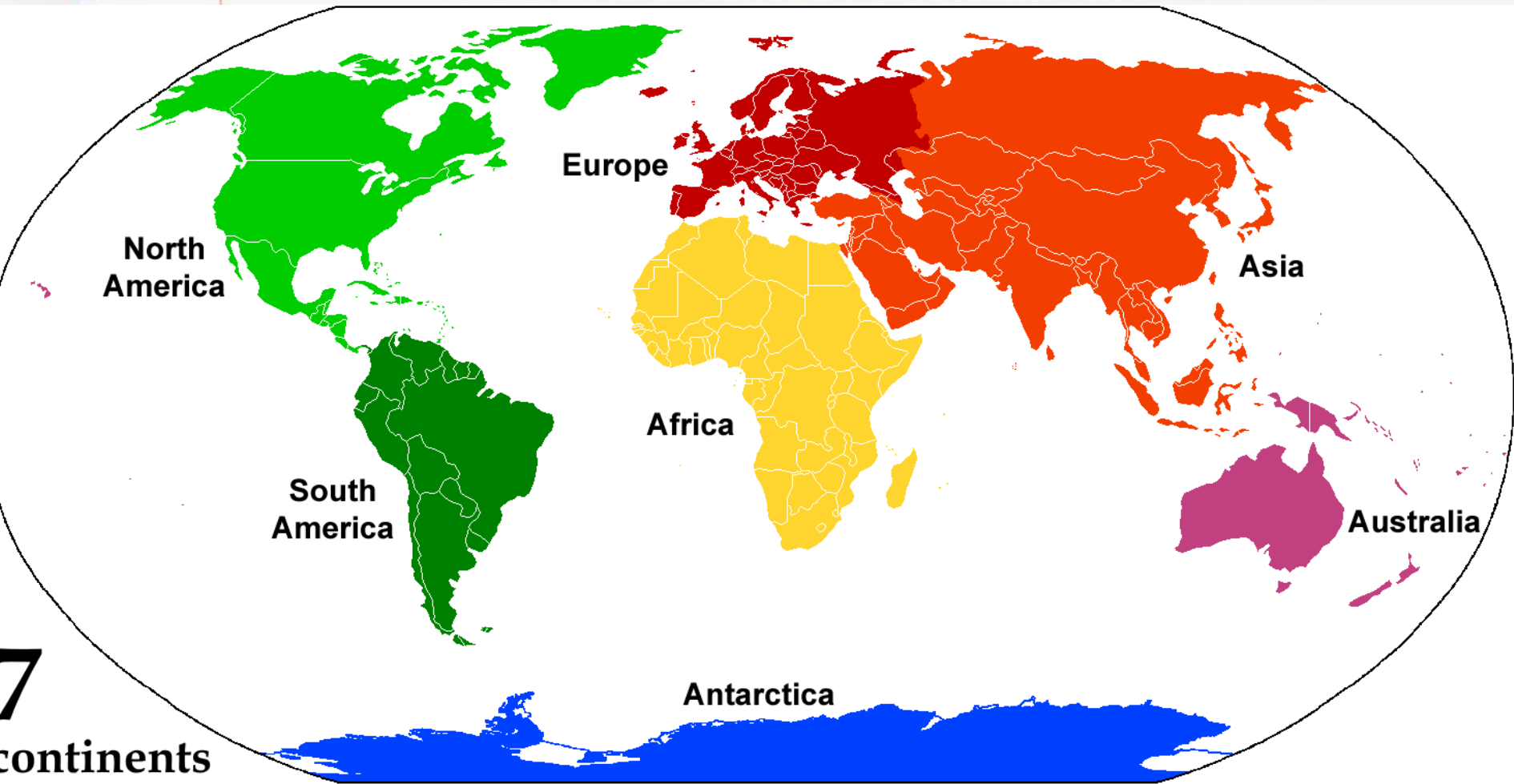
05:00
MR. TIMER

In "Mental World Map" section of your guided notes:

Visualize a map of the world

1. Draw your mental map of the world on your sheet of paper
 - a. Outline of the continents - label them
 - b. Outline border of the United States
 - c. Add the Equator & Prime Meridian
2. Can you add anything else?

CONTINENTS

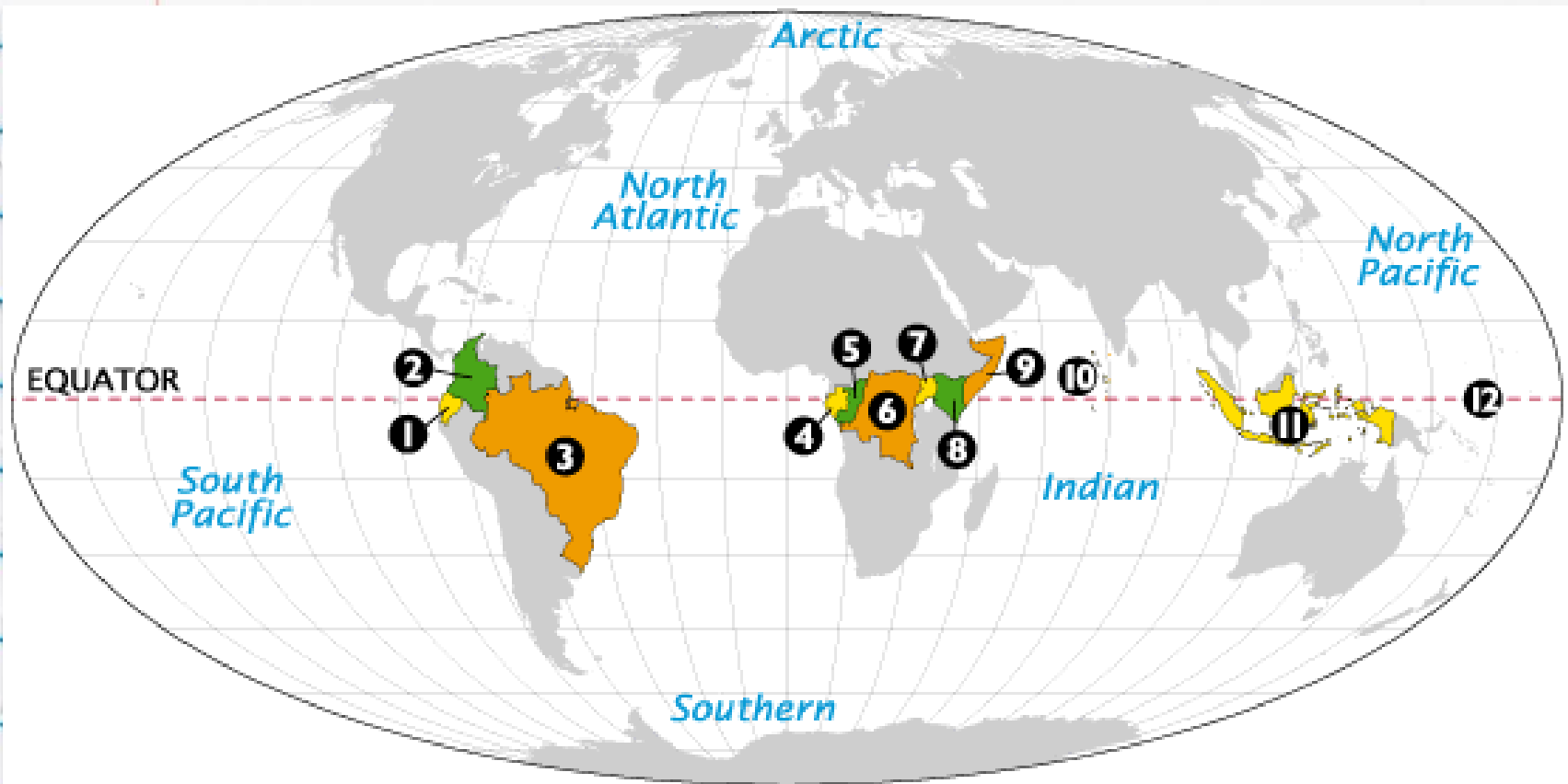


USA



©<https://worldmapwithcountries.net/>

ANALYZING YOUR WORLD: EQUATOR



❶ Ecuador

❷ Colombia

❸ Brazil

❹ Gabon

❺ Congo

❻ Dem. Rep. of Congo

❼ Uganda

❽ Kenya

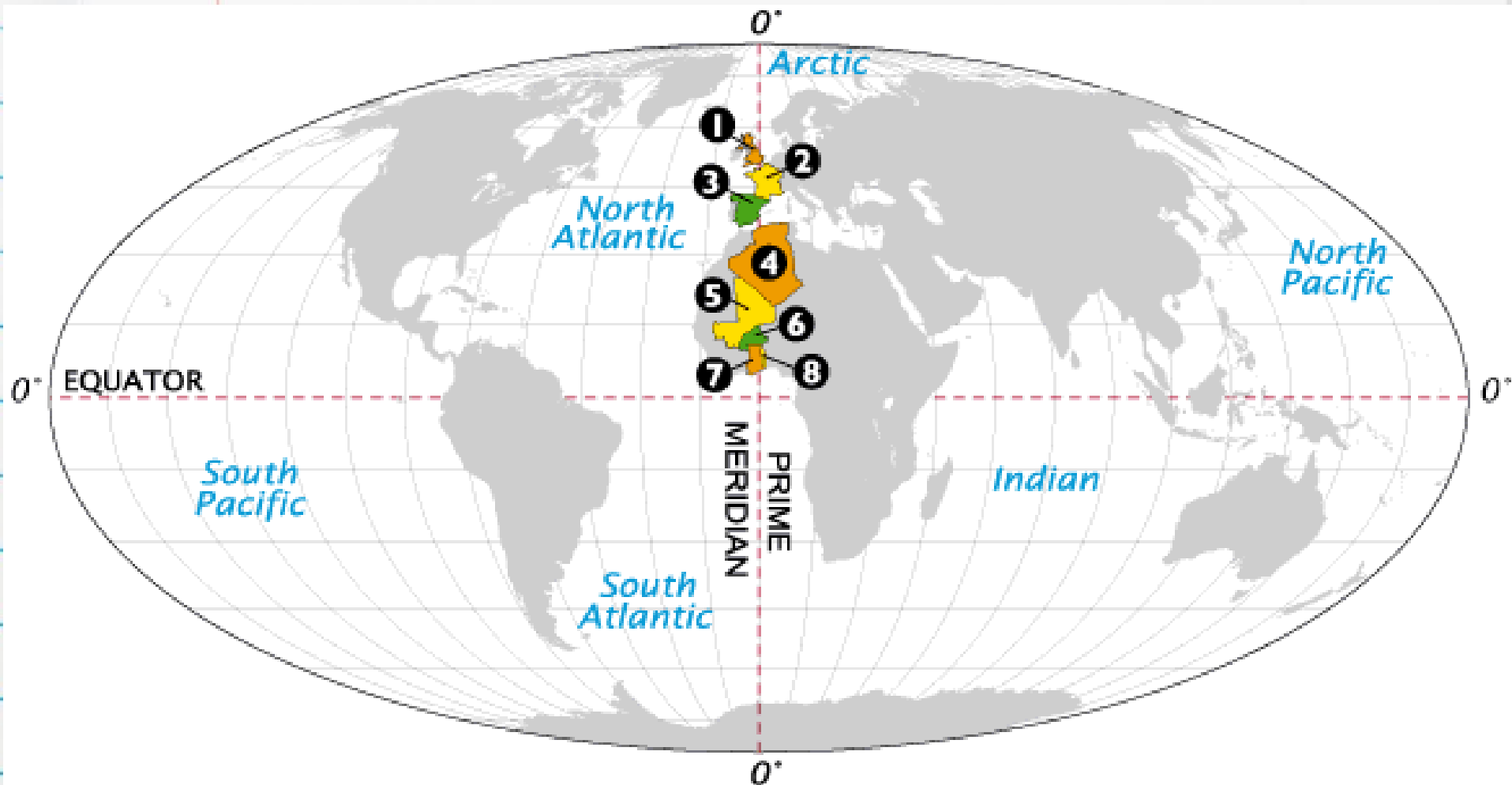
❾ Somalia

❿ Maldives

⓫ Indonesia

⓬ Kiribati

ANALYZING YOUR WORLD: PRIME MERIDIAN



- | | | | |
|------------------|-----------|---------|-----------|
| ① United Kingdom | ② France | ③ Spain | ④ Algeria |
| ⑤ Mali | ⑥ Burkina | ⑦ Ghana | ⑧ Togo |

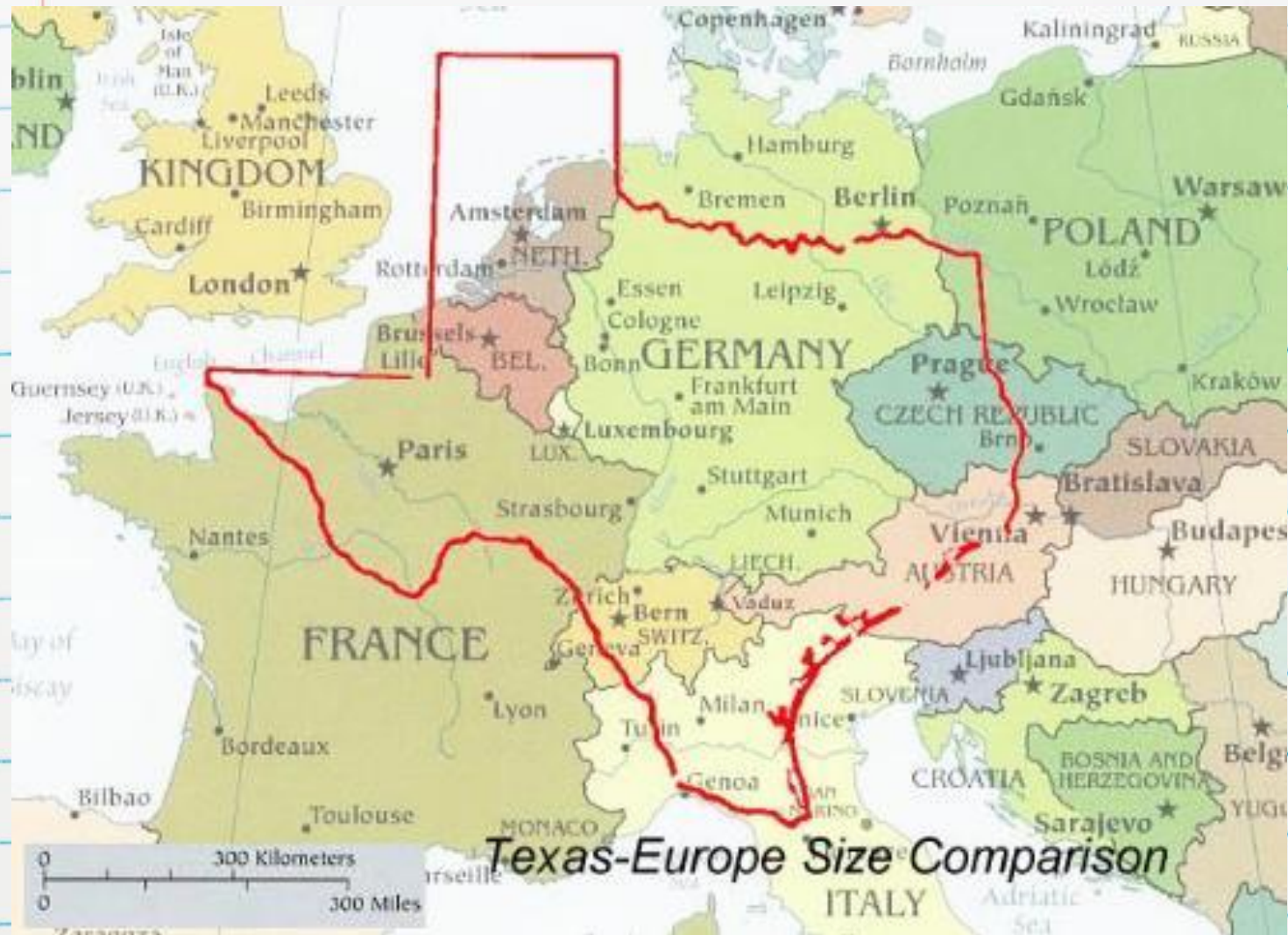
WORLD



ANALYZING YOUR WORLD: THE USA TAKES ON THE WORLD



ANALYZING YOUR WORLD: THE USA TAKES ON THE WORLD



The True Size of Africa

A small contribution in the fight against rampant *Immappancy*, by Kai Krause

Graphic layout for visualization only (some countries are cut and rotated)
But the conclusions are very accurate: refer to table below for exact data

COUNTRY	AREA x 1000 km ²
China	9.597
USA	9.629
India	3.287
Mexico	1.964
Peru	1.285
France	633
Spain	506
Papua New Guinea	462
Sweden	441
Japan	378
Germany	357
Norway	324
Italy	301
New Zealand	270
United Kingdom	243
Nepal	147
Bangladesh	144
Greece	132
TOTAL	30.102
AFRICA	30.221

In addition to the well known social issues of *illiteracy* and *innumeracy*, there also should be such a concept as "*immappancy*", meaning *insufficient geographical knowledge*.

A survey with random American schoolkids let them guess the population and land area of their country. Not entirely unexpected, but still rather unsettling, the majority chose "*1-2 billion*" and "*largest in the world*", respectively.

Even with Asian and European college students, geographical estimates were often off by factors of 2-3. This is partly due to the highly distorted nature of the predominantly used mapping projections (such as *Mercator*).

A particularly extreme example is the worldwide misjudgement of the true size of *Africa*. This single image tries to embody the massive scale, which is larger than the *USA*, *China*, *India*, *Japan* and *all of Europe*.....combined!

Top 100 Countries

Area in square kilometers, Percentage of World Total
Sources: Britannica, Wikipedia, Almanac 2010

	AREA km ²	%	
1	Russia	17.098.242	11,50
2	Canada	9.984.670	6,70
3	China	9.596.961	6,40
4	United States	9.629.091	6,40
5	Brazil	8.514.877	5,70
6	Australia	7.692.024	5,20
7	India	3.287.263	2,30
8	Argentina	2.780.400	2,00
9	Kazakhstan	2.724.900	1,80
10	Sudan	2.505.813	1,70
11	Algeria	2.381.741	1,60
12	Congo	2.344.858	1,60
13	Greenland	2.166.086	1,50
14	Saudi Arabia	2.149.690	1,40
15	Mexico	1.964.375	1,30
16	Indonesia	1.860.360	1,30
17	Libya	1.759.540	1,20
18	Iran	1.628.750	1,10
19	Mongolia	1.564.100	1,10
20	Peru	1.285.216	0,86
21	Chad	1.284.000	0,86
22	Niger	1.267.000	0,85
23	Angola	1.246.700	0,85
24	Mali	1.240.192	0,83
25	South Africa	1.221.037	0,82
26	Colombia	1.141.748	0,76
27	Ethiopia	1.104.300	0,74
28	Bolivia	1.098.581	0,74
29	Mauritania	1.025.520	0,69
30	Egypt	1.002.000	0,67
31	Tanzania	945.087	0,63
32	Nigeria	923.768	0,62
33	Venezuela	912.050	0,61
34	Namibia	824.116	0,55
35	Mozambique	801.590	0,54
36	Pakistan	796.095	0,53
37	Turkey	783.562	0,53
38	Chile	756.102	0,51
39	Zambia	752.612	0,51
40	Myanmar	676.578	0,45
41	Afghanistan	652.090	0,44
42	Somalia	637.657	0,43
43	France	632.834	0,43
44	C. African Rep	622.984	0,42
45	Ukraine	603.500	0,41
46	Madagascar	587.041	0,39
47	Botswana	582.000	0,39
48	Kenya	580.367	0,39
49	Yemen	527.968	0,35
50	Thailand	513.120	0,34
51	Spain	505.992	0,34
52	Turkmenistan	488.100	0,33
53	Cameroon	475.442	0,32
54	Papua New Guinea	462.840	0,31
55	Uzbekistan	447.400	0,30
56	Morocco	446.550	0,30
57	Sweden	441.370	0,30
58	Iraq	438.317	0,29
59	Paraguay	406.752	0,27
60	Zimbabwe	390.757	0,26
61	Japan	377.930	0,25
62	Germany	357.114	0,24
63	Rep. of Congo	342.000	0,23
64	Finland	338.419	0,23
65	Vietnam	331.212	0,22
66	Malaysia	330.803	0,22
67	Norway	323.802	0,22
68	Côte d'Ivoire	322.463	0,22
69	Poland	312.685	0,21
70	Oman	309.500	0,21
71	Italy	301.336	0,20
72	Philippines	300.000	0,20
73	Burkina Faso	274.222	0,18
74	New Zealand	270.467	0,18
75	Gabon	267.668	0,18
76	Western Sahara	266.000	0,18
77	Ecuador	256.369	0,20
78	Guinea	245.857	0,17
79	United Kingdom	242.900	0,16
80	Uganda	241.038	0,16
81	Ghana	238.539	0,16
82	Romania	238.391	0,16
83	Laos	236.800	0,16
84	Guyana	214.969	0,14
85	Belarus	207.600	0,14
86	Kyrgyzstan	199.951	0,13
87	Senegal	196.722	0,13
88	Syria	185.180	0,12
89	Cambodia	181.035	0,12
90	Uruguay	176.215	0,12
91	Suriname	163.820	0,11
92	Tunisia	163.610	0,11
93	Nepal	147.181	0,10
94	Bangladesh	143.998	0,10
95	Tajikistan	143.100	0,10
96	Greece	131.957	0,09
97	Nicaragua	130.373	0,09
98	North Korea	120.538	0,08
99	Malawi	118.484	0,08
100	Eritrea	117.600	0,08
TOP 100 TOTAL		132.632.524	89,34

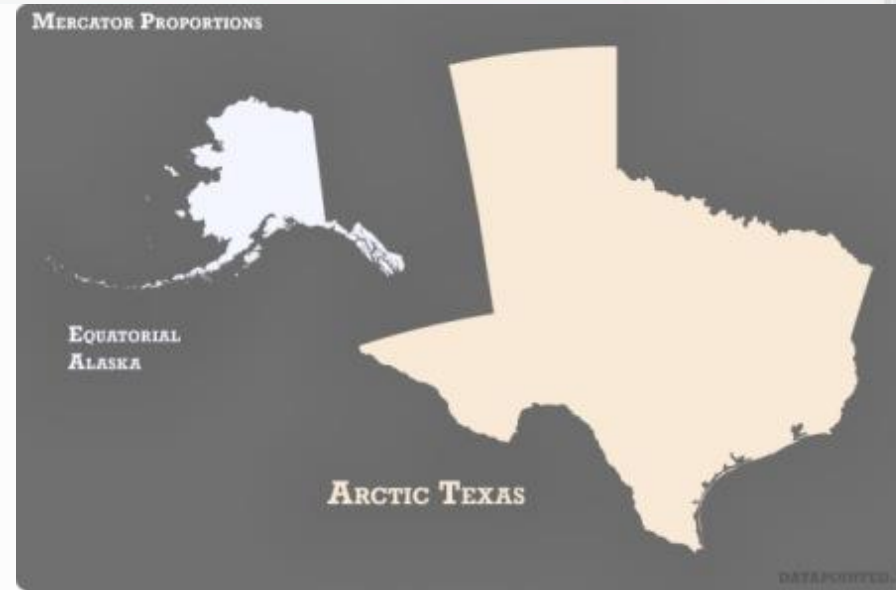


Why are our
perceptions of
sizes/shapes wrong?

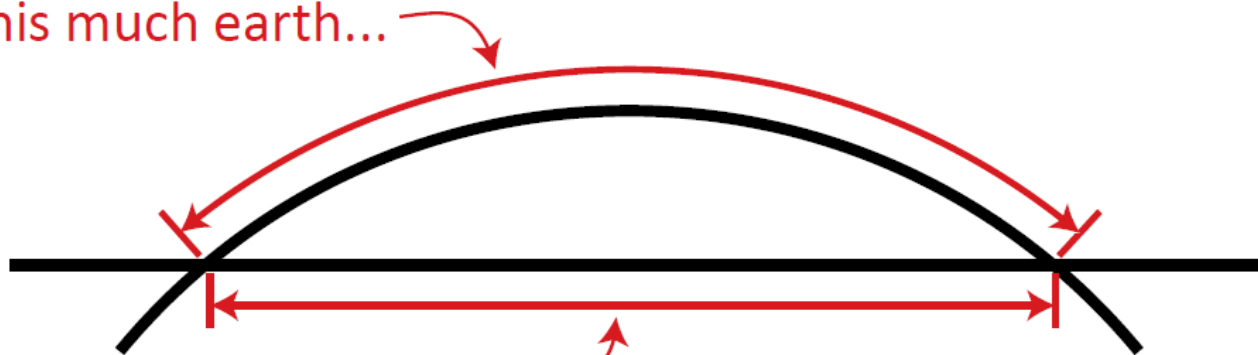
ALL MAPS ARE WRONG!!!!!!

Every two-dimensional (flat) map is *DISTORTED* in some way whether it is the shape, size, location, or direction of the map features

WHY DO MAPS APPEAR TO BE SO DISTORTED?



This much earth...

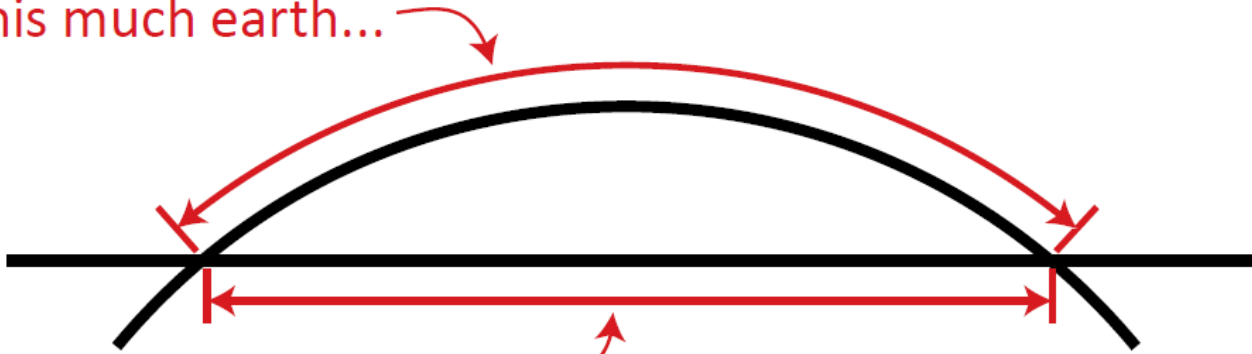


...has to fit onto this much map surface.

MAP PROJECTION

WHAT DOES IT ATTEMPT TO DO?

This much earth...



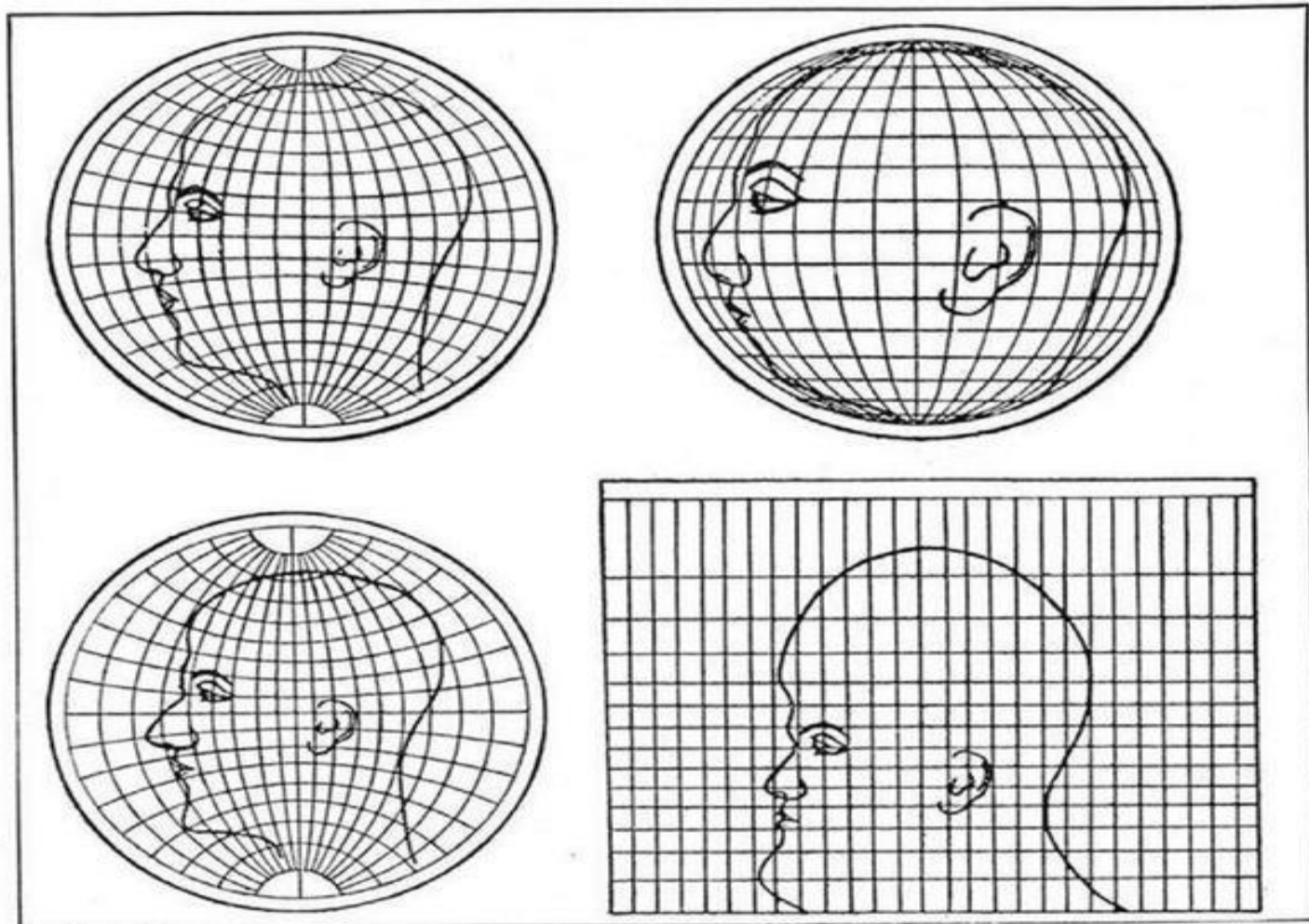
...has to fit onto this much map surface.

A TWO-DIMENSIONAL, OR FLAT, REPRESENTATION OF THE EARTH'S SURFACE.

WHY ALL MAPS ARE WRONG...



MAP PROJECTION DISTORTIONS: THE HUMAN HEAD



*Upper left: Globular. Upper right: Orthographic. Lower left: Stereographic.
Lower right: Mercator*

What four commonly used projections do, as shown on a human head

MAP PROJECTION DISTORTIONS: THE HUMAN HEAD



HOW CAN MAP DISTORTIONS IMPACT OUR PERCEPTIONS OF PLACES?

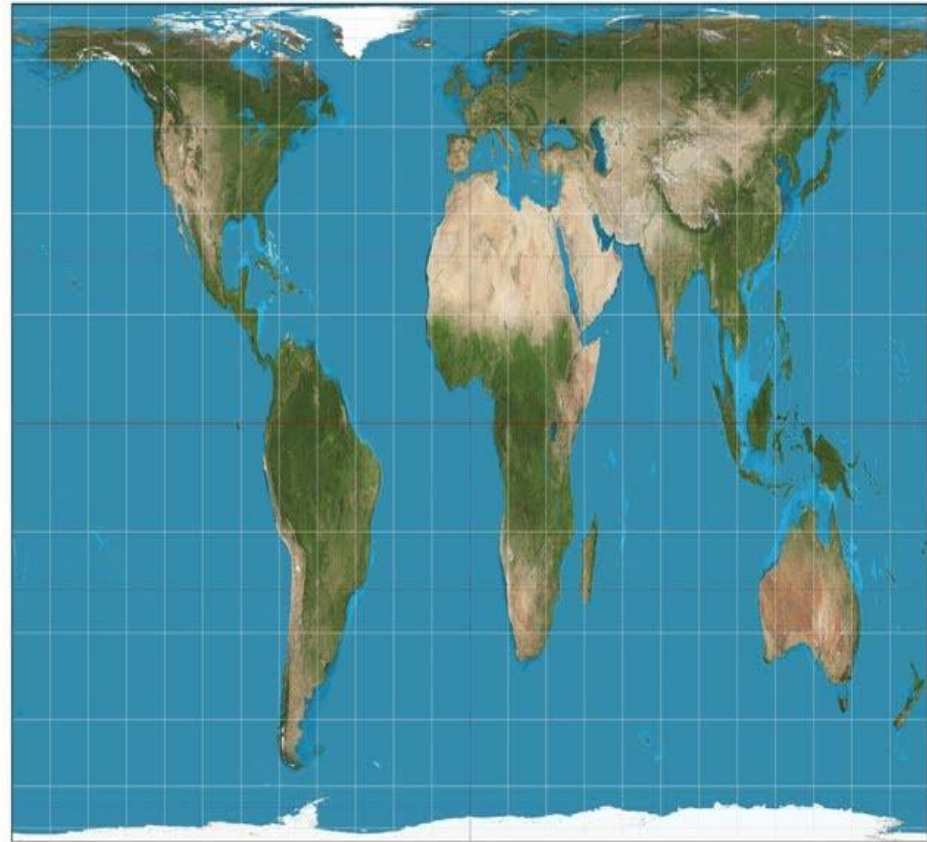


MERCATOR MAP VS. PETERS MAP

Which would you choose??



Mercator Projection



Gall-Peters Projection

EVEN THOUGH MAP PROJECTIONS ARE
WRONG, MAPS ARE STILL IMPORTANT AND
CAN BE USED TO DEPICT VARIOUS DATA

*Types of maps: Physical, Political, Reference,
and Thematic*

PHYSICAL MAPS

Show relief (change in elevation) and other physical land features



POLITICAL MAPS

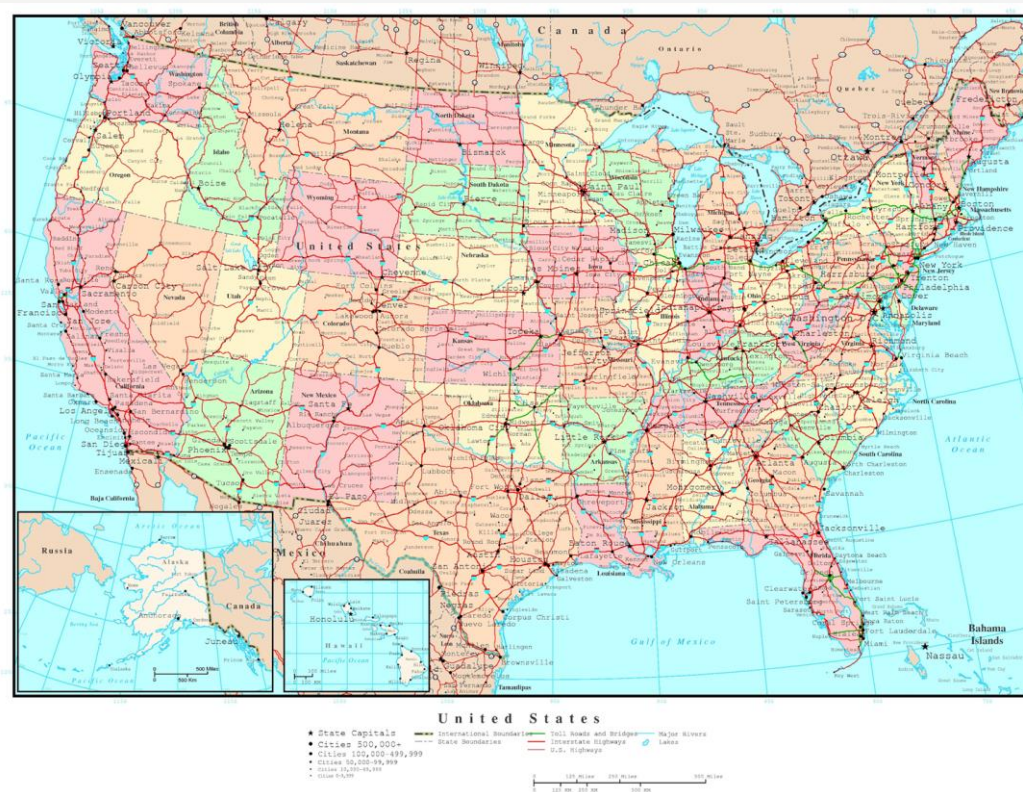
Show political boundaries (man-made borders and natural borders)



REFERENCE MAPS

General sources of information

Show the locations and names of places and geographic features

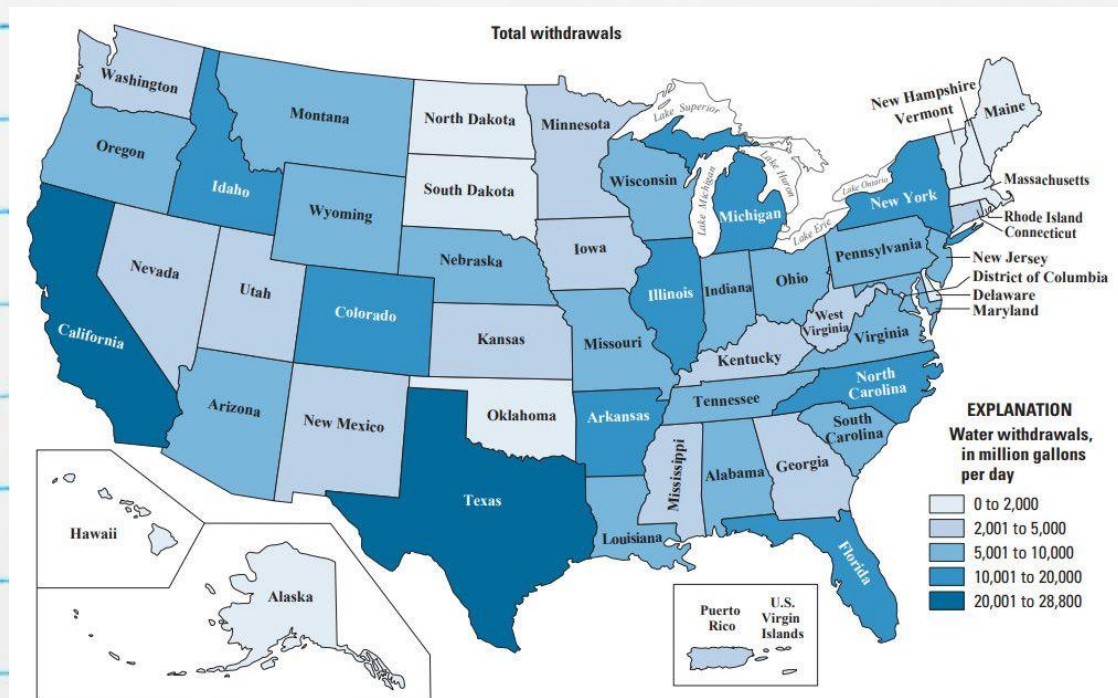


THEMATIC MAPS

Specific purpose or focus

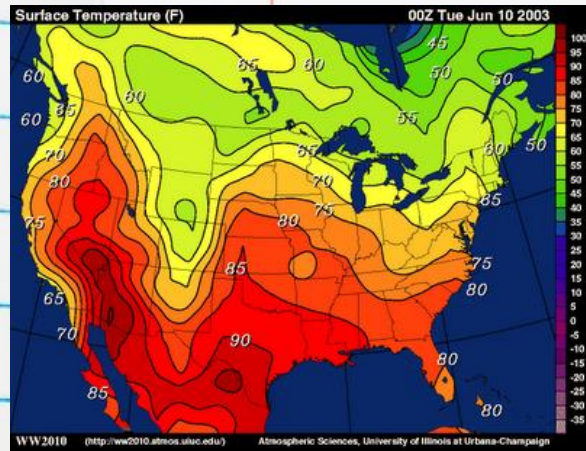
Show a particular theme connected to a geographic area - created for a special purpose

- tell a story

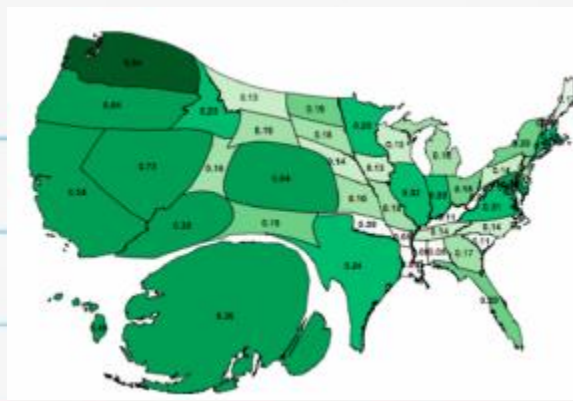
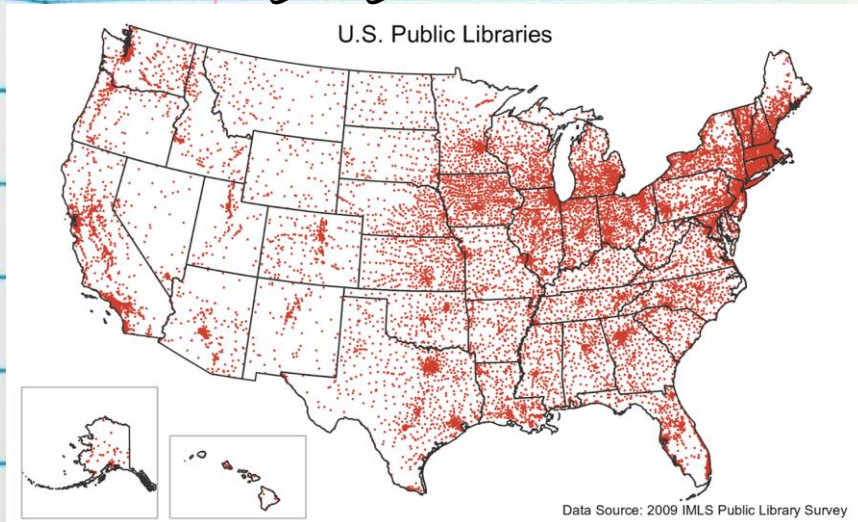


TYPES OF THEMATIC MAPS

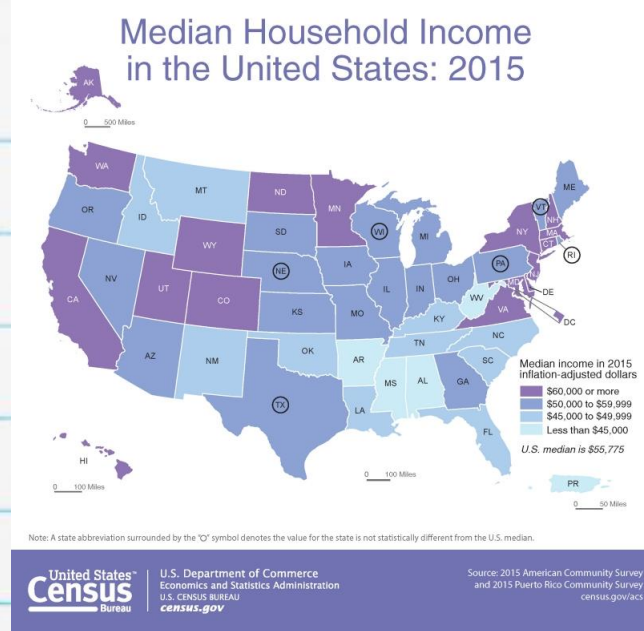
Isoline



Dot Distribution

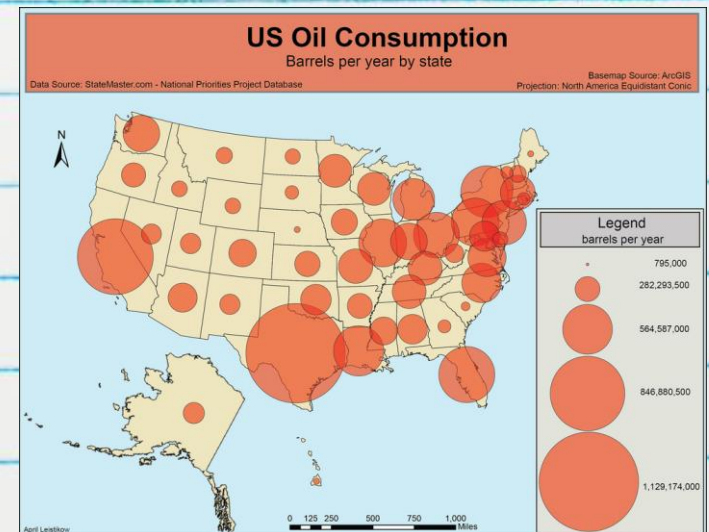


Cartogram



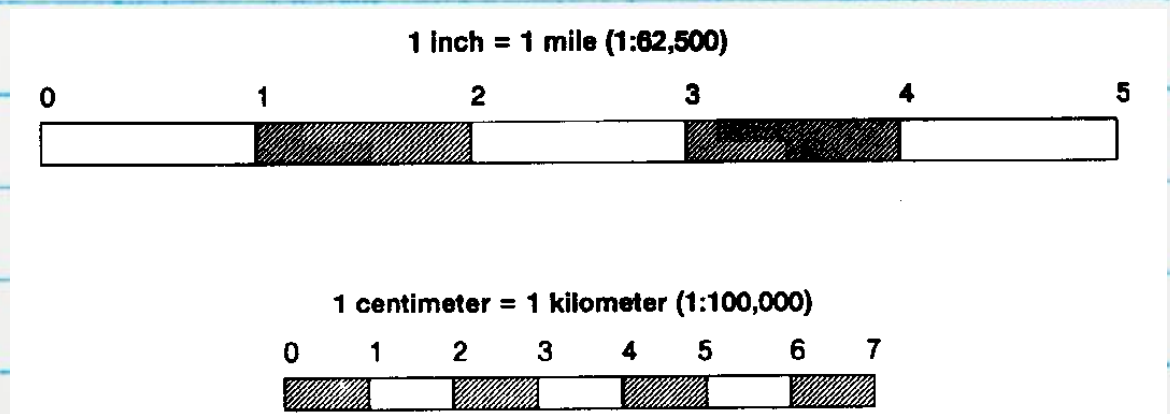
Choropleth

Graduated Symbol



MAP SCALE

- Distance on the map compares to distance on the ground
- The degree to which a map "zooms in" on the area



SMALLER SCALE

"Zoomed-out"



1 cm : 1,300 km
or
1 cm
130,000,000 cm
on the ground



1 cm : 960 km
or
1 cm
96,000,000 cm
on the ground



1 cm : 430 km
or
1 cm
43,000,000 cm
on the ground

LARGER SCALE

"Zoomed-in"



1 cm : 75 km
or
1 cm
7,500,000 cm
on the ground



Generally, large scale maps are ones that are most "zoomed-in", while small scale maps are most "zoomed-out".
Bottom line: Make sure to check the scale of a map before interpreting it's data.

WHY ARE MAPS IMPORTANT? WHY DO GEOGRAPHERS USE THEM?

- Help us to see spatial features and patterns - clustered, dispersed, or linear
 - Distribution of the phenomena
- Describe the spacing of people or places - absolute or relative distance
 - Absolute distance = measured using length (miles, Kilometers)
 - Relative distance = measured using other criteria (in relation to other places)
- Problem-solving
 - Ex. mapping the spread of a disease epidemic (Cholera map)

MAP GALLERY WALK PART I

Analyze each map with your group. Add to your handout:

1. The title of the map
2. Circle the type of map (you may circle more than one if appropriate)
3. Answer the question specific to the map you are analyzing -PAY ATTENTION TO MAP #1S ON WORKSHEET

SKIP THE QUESTION REGARDING THE SCALE OF ANALYSIS

GALLERY WALK GROUPS: STARTING POINT

Desks numbered 1-5: Doc A/1

Desks numbered 6-10: Doc B/2

Desks numbered 11-15: Doc C/3

Desks numbered 16-20: Doc D/4

Desks numbered 21-26: Doc E/5

Desks numbered 27-32 Doc F/6



5:00

REVIEW VIDEO